

Global Hilbert expansion for some non-relativistic kinetic equations

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Hilbert expansion (1912)

In 1912, Hilbert proposed the Hilbert expansion ([the first result in Hilbert's sixth problem](#)):

Boltzmann equation $\Rightarrow$ compressible Euler equations.

Namely,

$$\begin{cases} \partial_t F^\varepsilon + v \cdot \nabla_x F^\varepsilon = \frac{1}{\varepsilon} Q(F^\varepsilon, F^\varepsilon), \\ F^\varepsilon = \sum_{n=0}^{\infty} \varepsilon^n F_n, \quad (\varepsilon : \text{the Knudsen number}). \end{cases} \Rightarrow$$

$$\begin{cases} \varepsilon^{-1} : & Q(F_0, F_0) = 0, \\ \varepsilon^0 : & \partial_t F_0 + v \cdot \nabla_x F_0 = Q(F_0, F_1) + Q(F_1, F_0). \end{cases} \Rightarrow$$

$$\begin{cases} F_0 := \mathbf{M} = \frac{\rho}{(2\pi T)^{3/2}} \exp \left\{ -\frac{|v-u|^2}{2T} \right\}, \\ (\rho, u, T) \text{ satisfies the compressible Euler equations.} \end{cases}$$

Breakthrough by Caflisch

The first breakthrough was made by Caflisch (CPAM-1980) with a **truncated Hilbert expansion**:

$$F^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n F_n + \varepsilon^k F_R^\varepsilon, \quad k \geq 2.$$

- ◆ Coefficients $F_n (0 \leq n \leq 2k - 2)$: can be solved successively based on (ρ, u, T) .
- ◆ Main difficulty in solving the remainder term $F_R^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon$:

$$f^\varepsilon(t, x, v) \mathbf{M}^{-\frac{1}{2}}(t, x, v) \left\{ \partial_t + v \cdot \nabla_x \right\} \mathbf{M}^{\frac{1}{2}}(t, x, v) \sim \langle v \rangle^3 f^\varepsilon(t, x, v) \quad (1)$$

Breakthrough by Caflisch

♦ Via Caflisch's decomposition :

$$F_R^\varepsilon = \underbrace{\mathbf{M}^{\frac{1}{2}} f^\varepsilon}_{\text{low velocity part}} + \underbrace{\mu^{\frac{1}{2}} h^\varepsilon}_{\text{high velocity part}},$$

where

$$\mu(v) = \frac{1}{(2\pi T_c)^{3/2}} \exp\left\{-\frac{|v|^2}{2T_c}\right\}, \quad \frac{1}{2}T(t, x) < T_c < T(t, x),$$

F_R^ε was solved as a system

$$\begin{cases} (\partial_t + v \cdot \nabla_x) f^\varepsilon = F(f^\varepsilon, h^\varepsilon), \\ (\partial_t + v \cdot \nabla_x) h^\varepsilon = G(f^\varepsilon, h^\varepsilon), \\ f^\varepsilon(0, x, v) = h^\varepsilon(0, x, v) = 0. \end{cases}$$

♦ Drawback: $F^\varepsilon(t, x, v) \geq 0$ can not be guaranteed.

Methods to study the Hilbert expansion

L^2 - L^∞ method (inspired by Guo-ARMA-2010):
first introduced in Guo-Jang-Jiang (KRM-2009, CPAM-2010).

♦ **Main ideas:** interplay estimates of L^2 -norm and L^∞ -norm.

♦ **Key estimates:**

$$\left\langle \left| \mathbf{M}^{-\frac{1}{2}} (\partial_t + v \cdot \nabla_x) \mathbf{M}^{\frac{1}{2}} \right| f^\varepsilon, f^\varepsilon \right\rangle \lesssim \|\nabla_{x,t}(\rho, u, T)\| \|h^\varepsilon\|_\infty \|f^\varepsilon\|,$$
$$\|h^\varepsilon\|_\infty \lesssim \varepsilon^{\frac{3}{2}} \|f^\varepsilon\| + \dots, \dots,$$

where $\beta > \frac{9-2\gamma}{2}$ and h^ε satisfies

$$(1 + |v|)^{-\beta} \mu^{\frac{1}{2}} h^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = F_R^\varepsilon, \quad \frac{1}{2} T(t, x) < T_c < T(t, x).$$

Methods to study the Hilbert expansion

◆ Further developments:

- VPB system: Guo-Jang-CMP-2010
- r-Boltzmann equation: Speck-Strain-CMP-2011
- r-VMB system: Guo-Xiao-CMP-2021

Remark: Since the L^2 – L^∞ method heavily relies on the characteristic method, it seems difficult to be applied in the Landau type and non-cutoff Boltzmann type equations.

Problem: What about the Hilbert expansion of Landau type and non-cutoff Boltzmann type equations?

Methods to study the Hilbert expansion

Exponentially weighted energy method (inspired by Duan-Yang-Zhao (JDE-2012, M3AS-2013)):

- first introduced in Ouyang-Wu-Xiao (arXiv:2207.00126)
- further development in Lei-Liu-Xiao-Zhao (arXiv:2209.15201, arXiv:2310.11745)

♦ **Main ideas:**

- overcome the velocity growth via a special weight function
- interplay energy estimates involving local and global Maxwellians

♦ **Key estimates:** For $w_e = \exp \left\{ \frac{\bar{c}\langle v \rangle^a}{\ln(e+t)T_c} \right\}$, $\bar{c} > 0$, $1 \leq a \leq 2$,

$$F_R^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = \mu^{\frac{1}{2}} h^\varepsilon,$$

$$w_e^2 f^\varepsilon \partial_t f^\varepsilon = \frac{1}{2} \frac{\partial}{\partial t} (w_e f^\varepsilon)^2 + \frac{\bar{c}\langle v \rangle^a}{(e+t)[\ln(e+t)]^2} (w_e f^\varepsilon)^2,$$

$$\|\langle v \rangle^{3/2} f^\varepsilon\|^2 \lesssim \|f^\varepsilon\|^2 + \varepsilon^{-1} \|(\mathbf{I} - \mathbf{P}_M) [f^\varepsilon]\|_{\mathbf{D}}^2 + e^{-\frac{C}{\sqrt{\varepsilon}}} \|h^\varepsilon\|^2.$$

Concerned systems

Consider the scaled Vlasov-Maxwell Landau system or non-cutoff Vlasov-Maxwell-Boltzmann system:

$$\left\{ \begin{array}{l} \partial_t F^\varepsilon + v \cdot \nabla_x F^\varepsilon - e(E^\varepsilon + v \times B^\varepsilon) \cdot \nabla_v F^\varepsilon = \frac{1}{\varepsilon} C[F^\varepsilon, F^\varepsilon], \\ \partial_t E^\varepsilon - c \nabla_x \times B^\varepsilon = 4\pi e \int_{\mathbb{R}^3} v F^\varepsilon dv, \\ \partial_t B^\varepsilon + c \nabla_x \times E^\varepsilon = 0, \\ \nabla_x \cdot E^\varepsilon = 4\pi e \left(1 - \int_{\mathbb{R}^3} F^\varepsilon dv\right), \\ \nabla_x \cdot B^\varepsilon = 0. \end{array} \right.$$

- $F^\varepsilon(t, x, v)$: distribution function for electrons
- $-e$: electrons' charge
- $C[\cdot, \cdot]$: Landau (Boltzmann) collision operator
- $E^\varepsilon, B^\varepsilon$: electromagnetic field

Corresponding to hydrodynamic level model, Euler-Maxwell system:

$$\begin{cases} \partial_t \rho + \nabla_x \cdot (\rho u) = 0, \\ \partial_t [\rho u] + \nabla_x \cdot (\rho u \otimes u) + \frac{2}{3} \nabla_x \rho^{\frac{5}{3}} + \rho (E + u \times B) = \mathbf{0}, \end{cases}$$

$$\begin{cases} \partial_t E - c \nabla_x \times B = 4\pi e n u, \\ \partial_t B + c \nabla_x \times E = \mathbf{0}, \\ \nabla_x \cdot E = 4\pi e (1 - n), \quad \nabla_x \cdot B = 0. \end{cases}$$

Theorem(Ionescu-Pausader-JEMS-2014):

Let $(\rho(0, x), u(0, x), T(0, x), E(0, x), B(0, x))$ be irrotational and be close to constant state $(1, \mathbf{0}, 1, \mathbf{0}, \mathbf{0})$. Then $\exists!$ global solution $(\rho(t, x), u(t, x), T(t, x), E(t, x), B(t, x))$ to the EM system that is irrotational for any $t > 0$ and it satisfies

$$\begin{aligned} & \sup_{t \in [0, \infty)} \|(\rho(t) - 1, u(t), T(t) - 1, E(t), B(t))\|_{H^{N_0}} \\ & + \sup_{t \in [0, \infty)} \sup_{|\rho| \leq \bar{N}} \left((1 + t)^{p_0} \left\| \nabla_x^\rho (\rho(t) - 1, u(t), T(t) - 1, E(t), B(t)) \right\|_{L^\infty} \right) \lesssim \bar{\varepsilon}_0, \end{aligned} \quad (2)$$

where $N_0 \in \mathbb{N}$ is sufficiently large, $3 \leq N_1 \ll N_0$, $p_0 = \frac{101}{100}$ and $\bar{\varepsilon}_0$ is sufficiently small.

Reference review

- ◆ Time-asymptotic equivalence of kinetic equations and fluid equations:
 - Boltzmann type equations
 - Landau type equations
- ◆ Parameter (Knudsen number, Strouhal number) limits of kinetic equations:
 - Compressible Euler limit
 - Incompressible Euler limit
 - Incompressible Navier-Stokes-Fourier limit
 - Compressible Navier-Stokes approximation
 - Hilbert expansion

Bardos, Caflish, Duan, Golse, Guo, Huang, Jang, Jiang, Levermore, Liu, Masmoudi, Nishida, Saint-Raymond, Strain, Ukai, Wang, Yang, Yu, Zhao

.....

Hilbert expansion of the VML system

Theorem (VML): Let $F^\varepsilon(0, x, p) \geq 0$. Assume that $(n(t, x), u(t, x), T(t, x), E(t, x), B(t, x))$ is the global solution constructed in Ionescu-Pausader-JEMS-2014. Then for $k \geq 3$ in the Hilbert expansion

$$F^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n F_n + \varepsilon^k F_R^\varepsilon, \quad E^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n E_n + \varepsilon^k E_R^\varepsilon, \quad B^\varepsilon = \sum_{n=0}^{2k-1} \varepsilon^n B_n + \varepsilon^k B_R^\varepsilon,$$

and $f^\varepsilon = \mathbf{M}^{-\frac{1}{2}} F_R^\varepsilon$, there exists an $\varepsilon_1 > 0$ such that for $0 \leq \varepsilon \leq \varepsilon_1$ and $0 < t \leq \bar{t}$ with $\bar{t} = \varepsilon^{-1/3}$, if $\frac{\varepsilon_1}{C_0} \leq T(t, x) - T_c \leq C_0 \varepsilon_1$,

$$\bar{\mathcal{E}}(0) \lesssim 1,$$

The VML system admits a unique solution satisfying $F^\varepsilon(t, x, p) \geq 0$ and

$$\sup_{0 \leq t \leq \bar{t}} \bar{\mathcal{E}}(t) + \int_0^{\bar{t}} \bar{\mathcal{D}}(s) ds \lesssim \bar{\mathcal{E}}(0) + 1. \quad (3)$$

Hilbert expansion of the VML system (continued)

where the energy functional $\bar{\mathcal{E}}(t)$ and dissipation rate functional $\bar{\mathcal{D}}(t)$ are defined as follows:

$$\bar{\mathcal{E}}(t) = \sum_{i=0}^2 \varepsilon^i \left[\left(\left\| \sqrt{4\pi RT} \nabla_x^i f^\varepsilon(t) \right\|^2 + \left\| \nabla_x^i E_R^\varepsilon(t) \right\|^2 + \left\| \nabla_x^i B_R^\varepsilon(t) \right\|^2 \right) + \varepsilon^{\frac{4}{3}} \left\| w_i \nabla_x^i h^\varepsilon(t) \right\|^2 \right],$$

$$\bar{\mathcal{D}}(t) \simeq \sum_{i=0}^2 \varepsilon^i \left[\frac{1}{\varepsilon} \left\| \nabla_x^i (\mathbf{I} - \mathbf{P}_M) [f^\varepsilon](t) \right\|_{\mathbf{D}}^2 + \varepsilon^{\frac{1}{3}} \left\| w_i \nabla_x^i h^\varepsilon(t) \right\|_{\mathbf{D}}^2 + \varepsilon^{\frac{4}{3}} Y(t) \left\| (1 + |\nu|) w_i \nabla_x^i h^\varepsilon(t) \right\|^2 \right],$$

where $f^\varepsilon = \mathbf{M}^{-\frac{1}{2}} F_R^\varepsilon$, $h^\varepsilon = \mu^{-\frac{1}{2}} F_R^\varepsilon$.

Remark:

- In the energy functional $\bar{\mathcal{E}}(t)$, compared with norm of $\nabla_x^i f^\varepsilon$, a coefficient $\varepsilon^{\frac{4}{3}}$ is added for the norm of $\nabla_x^i h^\varepsilon$ to remove the $\frac{1}{\varepsilon}$ singularity and time growth for $t \in [0, \varepsilon^{-\frac{1}{3}}]$.
- No macroscopic dissipation of f^ε or h^ε is included in $\bar{\mathcal{D}}(t)$.

Main ingredients for the VML system

Key point: an interplay of the energy estimates in the local and global Maxwellian frameworks.

(i) Estimate of the velocity growth term (1):

Let $F_R^\varepsilon = \mathbf{M}^{\frac{1}{2}} f^\varepsilon = \mu^{\frac{1}{2}} h^\varepsilon$, we have

$$|\partial^\alpha f^\varepsilon| \lesssim \sum_{j=0}^{|\alpha|} \langle v \rangle^{2(|\alpha|-j)} \exp\left(-\frac{(T - T_c)|v|^2}{8RTT_c}\right) |\nabla_x^j h^\varepsilon|, \quad 0 \leq |\alpha| \leq 2.$$

Take $|\alpha| = 0$ and use $\frac{\varepsilon_1}{C_0} \leq T(t, x) - T_c \leq C_0 \varepsilon_1$ to have

$$\begin{aligned} & \|\langle v \rangle^{3/2} f^\varepsilon\|^2 \\ & \lesssim \|\langle v \rangle^{3/2} \mathbf{P}_M[f^\varepsilon]\|^2 + \|\langle v \rangle^{3/2} (\mathbf{I} - \mathbf{P}_M)[f^\varepsilon]\|^2 \\ & \lesssim \|f^\varepsilon\|^2 + \int_{\mathbb{R}^3} \int_{\langle v \rangle^4 \leq \varepsilon^{-1}} \langle v \rangle^3 |(\mathbf{I} - \mathbf{P}_M)[f^\varepsilon]|^2 dx dv + \int_{\mathbb{R}^3} \int_{\langle v \rangle^4 \geq \varepsilon^{-1}} \langle v \rangle^3 |f^\varepsilon|^2 dx dv \\ & \lesssim \|f^\varepsilon\|^2 + \varepsilon^{-1} \|(\mathbf{I} - \mathbf{P}_M)[f^\varepsilon]\|_{\mathbf{D}}^2 + C(\varepsilon_1) e^{-\frac{\varepsilon_1}{8C_0RT_c^2\sqrt{\varepsilon}}} \|h^\varepsilon\|^2. \end{aligned}$$

(ii) Estimate of the term $\frac{1}{\varepsilon} \mathcal{L}_{\mathbf{D}}[h^\varepsilon]$:

Note that

$$-\mathcal{L}_{\mathbf{D}}[h^\varepsilon] = \mu^{-\frac{1}{2}} \left[C(\mathbf{M} - \mu, \mu^{\frac{1}{2}} h^\varepsilon) + C(\mu^{\frac{1}{2}} h^\varepsilon, \mathbf{M} - \mu) \right].$$

Since $\mathbf{M}$ and μ are sufficiently close, we have

$$\frac{1}{\varepsilon} |\langle \mathcal{L}_{\mathbf{D}}[h^\varepsilon], h^\varepsilon \rangle| \lesssim \frac{\epsilon_1}{\varepsilon} \|h^\varepsilon\|_{\mathbf{D}}^2 \lesssim \frac{\epsilon_1}{\varepsilon} \left(\|(\mathbf{I} - \mathbf{P})[h^\varepsilon]\|_{\mathbf{D}}^2 + \|\mathbf{P}[h^\varepsilon]\|_{\mathbf{D}}^2 \right).$$

On the other hand, we have

$$\frac{\epsilon_1}{\varepsilon} \|\mathbf{P}[h^\varepsilon]\|_{\mathbf{D}}^2 \lesssim \frac{\epsilon_1}{\varepsilon} \|\mathbf{P}_{\mathbf{M}}[f^\varepsilon]\|^2 \lesssim \frac{\epsilon_1}{\varepsilon} \|f^\varepsilon\|^2.$$

To make the red term time-integrable over $[0, \varepsilon^{-1/3}]$, we add additional weight $\varepsilon^{4/3}$ to the norm $\|h^\varepsilon\|^2$.

Main ingredients for the VML system

(iii) Design of ε -dependent norms in $\mathcal{E}(t)$:

$$\begin{aligned} \left\langle \frac{1}{\varepsilon} \nabla_x \mathcal{L}[f^\varepsilon], \nabla_x f^\varepsilon \right\rangle &= \left\langle \frac{1}{\varepsilon} \mathcal{L} \left[\nabla_x (\mathbf{I} - \mathbf{P})[f^\varepsilon] \right], (\mathbf{I} - \mathbf{P}) \nabla_x [f^\varepsilon] \right\rangle \\ &\quad - \left\langle \frac{1}{\varepsilon} \llbracket \mathcal{L}, \nabla_x \rrbracket (\mathbf{I} - \mathbf{P})[f^\varepsilon], \nabla_x f^\varepsilon \right\rangle \\ &\geq \frac{\delta}{\varepsilon} \left\| \nabla_x (\mathbf{I} - \mathbf{P})[f^\varepsilon] \right\|_\sigma^2 - \frac{C\mathcal{Z}}{\varepsilon^2} \left\| (\mathbf{I} - \mathbf{P})[f^\varepsilon] \right\|_\sigma^2 - C\mathcal{Z} \|f^\varepsilon\|_{H^1}^2, \end{aligned}$$

where $\mathcal{Z} = \sup_{x \in \mathbb{R}^3, 0 \leq \ell \leq 2} \left\{ \left| \nabla_{t,x}^{1+\ell}(n_0, u, T) \right| + \left| \nabla_{t,x}^\ell(E_0, B_0) \right| + \left| \nabla_{t,x}^\ell u \right| \right\}$.

To control the red terms, we design $\varepsilon \left\| \nabla_x f^\varepsilon \right\|^2$ in $\mathcal{E}(t)$. Similarly, $\varepsilon^2 \left\| \nabla_x^2 f^\varepsilon \right\|^2$ is also designed for the estimate of $\nabla_x^2 f^\varepsilon$.

Main ingredients for the VML system

(iv) time-dependent exponential weight function:

To proceed energy estimates of h^ε , introduce a time-related exponential function

$$w_{|\alpha|}(t, v) := \langle v \rangle^{\ell-|\alpha|} \exp\left(\frac{(1 + |v|^2)}{8RT_c \ln(e + t)}\right), \quad 0 \leq |\alpha| \leq s < \ell,$$

where $|\alpha|$ is the order of x -derivatives.

- The exponential part is mainly designed to control velocity growth term such as $\frac{E \cdot v}{2RT_c} h^\varepsilon$
- The algebraic part is mainly designed to bound terms with v -derivatives like $(E + v \times B) \cdot \nabla_v h^\varepsilon$.

Remark: The weight $w_{|\alpha|}$ also leads to velocity growth in estimating $\varepsilon^{k-1} \Gamma(h^\varepsilon, h^\varepsilon)$. This problem is solved by the inequality:

$$\sup_{v \in \mathbb{R}^3} \left[\langle v \rangle^i \exp\left(\frac{-\epsilon_1 \langle v \rangle^2}{8C_0 RT_c \ln(e + t)}\right) \right] \lesssim \left(\frac{\ln(e + t)}{\epsilon_1} \right)^{\frac{i}{2}}.$$

Hilbert expansion of the non-cutoff VMB system

Additional difficulties compared to the VML system:

- weak dissipation of $\mathcal{L}_{\mathbf{M}}[f]$ or $(\mathcal{L}[f])$: absence of $\nabla_v f$;
 $\gamma + 2 \rightarrow -1$ for the Landau case and $\gamma + 2s \rightarrow -\frac{3}{2}$ for the non-cutoff Boltzmann case
- no estimate with weight $e^{\frac{q}{2}\langle v \rangle^2}$ ($0 < q < 1$) available (only $e^{\frac{q}{2}\langle v \rangle}$ ($0 < q < \infty$) type weight doable)

Main ingredients for the VMB system

Introduce velocity weight functions $w_1(w_2)$ for the VMB system without (with) cutoff: $w_i(t, v) := \exp\left(\frac{\langle v \rangle^i}{8RT_c \ln(e+t)}\right)$.

(i) Estimate of the nonlinear collision operator (without cutoff):

$$\begin{aligned}
 & \left| \left\langle w_1^2 \partial_\beta^\alpha \Gamma(g_1, g_2), \partial_\beta^\alpha g_3 \right\rangle \right| \\
 & \lesssim \sum \left\{ \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{s+\gamma/2} \left| \partial_{\beta_2}^{\alpha_2} g_2 \right|_{\mathbf{D}}, \left| \partial_{\beta_2}^{\alpha_2} g_2 \right|_{s+\gamma/2} \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{\mathbf{D}} \right\} \left| w_1 \partial_\beta^\alpha g_3 \right|_{\mathbf{D}} \\
 & + \min \left\{ \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{L^2} \left| \partial_{\beta_2}^{\alpha_2} g_2 \right|_{s+\gamma/2} + \left| \partial_{\beta_2}^{\alpha_2} g_2 \right|_{L^2} \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{s+\gamma/2} \right\} \left| w_1 \partial_\beta^\alpha g_3 \right|_{\mathbf{D}} \\
 & + \sum \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{s+\gamma/2} \left| e^{\frac{\langle v \rangle}{8RT_c \ln(e+t)}} \partial_{\beta_2}^{\alpha_2} g_2 \right|_{L^2} \left| w_1 \partial_\beta^\alpha g_3 \right|_{\mathbf{D}} \\
 & + \sum \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{1/2+\gamma/2} \left| \mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2 \right|_{L^2} \left| w \partial_\beta^\alpha g_3 \right|_{1/2+\gamma/2} \\
 & + \sum [\ln(1+t)]^{-1} \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{1+\gamma/2} \left| \mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2 \right|_{L^2} \left| w_1 \partial_\beta^\alpha g_3 \right|_{s+\gamma/2} \\
 & + \sum [\ln(1+t)]^{-2} \left| w_1 \partial_{\beta_1}^{\alpha_1} g_1 \right|_{1+\gamma/2} \left| \mu^{\frac{\lambda_0}{128}} \partial_{\beta_2}^{\alpha_2} g_2 \right|_{L^2} \left| w_1 \partial_\beta^\alpha g_3 \right|_{1+\gamma/2},
 \end{aligned}$$

(ii) Estimate of the transport term $v \cdot \nabla_x h^\varepsilon$:

Noting that for $|\beta| \geq 1$,

$$|\langle \partial_\beta^\alpha (v \cdot \nabla_x h^\varepsilon), w_i^2 \partial_\beta^\alpha h^\varepsilon \rangle| \lesssim \|w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon\| \|w_i \partial_\beta^\alpha h^\varepsilon\|,$$

we can obtain

$$\begin{aligned} & |\langle \partial_\beta^\alpha (v \cdot \nabla_x h^\varepsilon), w_i^2 \partial_\beta^\alpha h^\varepsilon \rangle| \\ & \lesssim \varepsilon^{\frac{1}{6}} \left(\frac{1}{\varepsilon} \|w_i \partial_\beta^\alpha h^\varepsilon\|_{\mathbf{D}}^2 + Y(t) \|\langle v \rangle^{\frac{i}{2}} w_i \partial_\beta^\alpha h^\varepsilon\|^2 \right. \\ & \quad \left. + \frac{1}{\varepsilon} \|w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon\|_{\mathbf{D}}^2 + Y(t) \|\langle v \rangle^{\frac{i}{2}} w_i \nabla_x^{|\alpha|+1} \nabla_v^{|\beta|-1} h^\varepsilon\|^2 \right). \end{aligned}$$

(iii) Estimates of the nonlinear terms:

$$\frac{E \cdot v}{2RT_c} h^\varepsilon, \quad (E + v \times B) \cdot \nabla_v h^\varepsilon$$

(iv) Construction of coefficients (F_n, E_n, B_n) , $1 \leq n \leq 2k - 1$
(without cutoff):

For $q \in (0, \infty)$, $\ell \geq 0$,

$$\left| \partial_t^{\alpha_0} \partial_\beta^\alpha \left(\mu^{-\frac{1}{2}} F_n \right) \right| \lesssim \langle v \rangle^{-\ell + n + |\gamma| + 2s/2} e^{-q\langle v \rangle} (1+t)^n. \quad (4)$$

Main difficulty to prove (4): $\frac{\mathbf{M}}{\mu} \sim e^{c_0 \langle v \rangle^2}$.

Key points to show (4):

- Set $F_n = \mathbf{M}^{\frac{1}{2}} f_n = \mu^{\frac{1}{2}} h_n$
- Solve F_n via f_n , while prove (4) via h_n

Thank you for your attentions!